

Seminar Task Description: Detecting Oil Slicks

Deep Learning for Maritime Vision Applications

This seminar task focuses on the detection of oil slicks in ocean Sentinel-1 SAR imagery. The motivation is that oil spills cause severe ecological damage and are expensive to monitor manually across large stretches of ocean. Synthetic aperture radar (SAR) is the sensor of choice for operational monitoring because it works at night and through clouds, and because oil films dampen short surface waves and therefore appear as dark patches in radar backscatter. The project investigates whether deep learning can be used to detect such slicks reliably and whether recent Geospatial Foundation Models (GFMs) improve over a CNN baseline.

The main research question is:

How well can deep learning detect oil slicks in Sentinel-1 SAR imagery, and how does performance change between an in-distribution random split and a geographic out-of-distribution split (Mediterranean)?

An optional extension question is:

Can a Geospatial Foundation Model improve over a CNN baseline, and what do XAI methods reveal about the features that drive the predictions?

Expected Scope of the Student Work

The core goal is to compare a CNN-from-scratch baseline with a Geospatial Foundation Model (GFM) variant on the WaterBench OilSlick task, restricted to the Sentinel-1 modality. This can include in particular:

- building a reproducible S1 data pipeline that downloads the relevant archives, extracts the GeoTIFFs, and loads them as VV/VH tensors,
- implementing a CNN baseline trained from scratch (e.g. ResNet-18 or ResNet-50 with a 2-channel input stem),
- implementing a GFM-based variant fine-tuned on the same S1 inputs (e.g. TerraMind; see Starter References for alternatives),
- evaluating both approaches on the **random** split and the **geographic** (Mediterranean OOD) split, using accuracy, F1 score, and AUROC,
- analysing failure cases and lookalikes (natural slicks, low-wind areas, ship wakes) using confusion matrices and example chips,
- carrying out explainability analysis, for example Grad-CAM for the CNN.

Provided Data

The dataset for this seminar task is the *OilSlick* subset of WaterBench (<https://huggingface.co/datasets/ayushprd/WaterBench>). It contains 1,363 binary-labelled chips at 224×224 pixels, with two Sentinel-1 polarisations channels (VV, VH) stored as GeoTIFFs.

Sample identifiers follow four prefixes:

- `pos_*` – positive (oil slick present), ≈ 500 samples,
- `neg_*` – negative (no oil slick), ≈ 500 samples,
- `ext_pos_*` – 233 extended positives,
- `ext_neg_*` – 130 extended negatives.

The extended (`ext_`) samples are used to construct the geographic out-of-distribution split.

Splits

Two splits are provided. The `random` split is in-distribution, the `geographic` split holds out the Mediterranean Sea as a spatial OOD region.

Split	Train	Validation	Test
random	900	150	300
geographic (Mediterranean OOD)	864	217	150

File Layout and Metadata

After extracting the S1 archives, images are organised as:

- `images_s1/{sample_id}_s1.tif` – 2-band GeoTIFF (VV, VH),
- `metadata.csv` / `metadata.json` – per-sample attributes,
- `splits/random/{train,val,test}.txt` – one sample id per line,
- `splits/geographic/{train,val,test}.txt` – spatial OOD split.

The metadata table contains 24 columns, grouped as:

- identity: `sample_id`, `label`, `subcategory`,
- geospatial: `center_lon`, `center_lat`, `proj_epsg`,
- quality: `nodata_fraction`, `is_valid`, `cloud_cover`,
- scene: `scene_id`, `platform`, `scene_datetime`, `source_date`, `date_diff_days`,
- statistics: `reflectance_min`, `reflectance_max`, `reflectance_mean`, `n_bands`, `band_order`.

Note that `cloud_cover`, `rgb_path`, and `swir_path` refer to the optional Sentinel-2 reference imagery and can be ignored when working only with S1.

Practical Notes

The OilSlick S1 dataset is approximately 7.5 GB and can be downloaded with:

```
hf download ayushprd/WaterBench --repo-type dataset \
  --include 'data/OilSlick/OilSlick-images_s1-00.tar' \
  'data/OilSlick/OilSlick-images_s1-01.tar' \
  'data/OilSlick/metadata.csv' \
  'data/OilSlick/metadata.json' \
  'data/OilSlick/splits/random/*.txt' \
  'data/OilSlick/splits/geographic/*.txt' \
  --local-dir <target-dir>
```

Then extract the archives:

```
cd <target-dir>/data/OilSlick && for f in *.tar; do tar xf "$f"; done
```

Baseline Approaches

CNN from scratch. A standard image-classification CNN such as ResNet-18 or ResNet-50 with a 2-channel input stem is a sensible baseline. Each chip should be normalised per channel (z-score on VV and VH backscatter in dB), and trained with binary cross-entropy. Note that U-Net or other segmentation architectures are not required: labels are image-level binary, not pixel masks.

Geospatial Foundation Models. A second variant replaces the encoder with a GFM that has been pretrained on Sentinel-1 SAR. The recommended primary choice is *TerraMind* (IBM / ESA), which natively accepts S1 GRD or RTC inputs. The model checkpoints are available on Hugging Face and can be fine-tuned with a small linear or MLP head for binary classification.

Explainability

To better understand the learned decision rules, XAI method can be applied. For the CNN baseline, Grad-CAM produces class-activation heatmaps that localise where the network is looking on the SAR chip. For either backbone, Integrated Gradients and SHAP (both available in the `captum` library) provide pixel-level attributions. Comparing attributions on positive samples (real slicks) and confidently misclassified lookalikes is a particularly informative analysis.

Starter References

- WaterBench dataset: <https://huggingface.co/datasets/ayushprd/WaterBench>
- TerraMind (Jakubik et al., 2025), arXiv:2504.11171, <https://huggingface.co/ibm-esa-geospatial>
- Grad-CAM (Selvaraju et al., 2017), arXiv:1610.02391
- Integrated Gradients (Sundararajan, Taly, Yan, 2017), arXiv:1703.01365
- Captum library: <https://captum.ai>

Supervisor

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